# Improving Trust in AI-Assisted Education through Inference-Based Factuality Verification

Subhajit Bag[1], Deepak Alagusubramanian[1], Aditya Ghosh[1], and Soham Pradhan[1]

[1]Georgia Institute of Technology, Atlanta, USA

## Abstract

Large language models are increasingly used in educational settings; however, they can produce fluent but incorrect responses, creating risks of misinformation at scale. This work addresses the societal challenge of verifying the factual correctness of AI-generated educational answers. We develop an analytics pipeline that reframes factuality verification as a natural language inference problem between generated answers and supporting context. Using transformer-based inference models with selective large language model arbitration for ambiguous cases, our approach achieves 99.11% balanced accuracy on a held-out test set, outperforming baseline classification methods. The results demonstrate that correct problem framing is critical for reliable AI oversight. By enabling scalable detection of factual errors, this work supports safer deployment of AI tools in education and contributes to broader efforts toward trustworthy and responsible artificial intelligence.

**Keywords:** factuality detection; natural language inference; trustworthy AI; education; model validation

## 1. Introduction

Artificial intelligence systems are rapidly becoming embedded in educational workflows, supporting students with explanations, problem-solving, and personalized learning assistance. According to the survey of [Campbell University Academic Technology, 2025] over 80% of university students have used AI tools in their studies, with more than half reporting weekly usage and a substantial fraction relying on AI-generated explanations as a primary learning aid rather than a supplementary resource . This marks a structural shift in how knowledge is accessed, interpreted, and internalized in formal education.

Despite these benefits, current AI systems exhibit a critical failure mode: the generation of fluent, confident, but factually incorrect responses. Empirical evaluations by [AI Multiple, 2024] across large language models consistently report hallucination or factual error rates exceeding 15% on structured reasoning and knowledge-intensive tasks, with error rates rising substantially for complex, multi-step, or domain-specific questions. Importantly, these failures are often not signaled by uncertainty or hedging language, making them difficult for learners to detect.

In educational settings, this failure mode is particularly harmful. Prior work by [Parker et al., 2026] in learning sciences shows that exposure to plausible but incorrect explanations can reinforce misconceptions, degrade conceptual understanding, and increase learner overconfidence, especially among novices who lack strong prior knowledge. At scale, such misinformation risks undermining learning outcomes, propagating false knowledge, and eroding trust in AI-assisted educational platforms. As AI increasingly mediates learning for millions of students, unchecked factual errors pose a systemic risk rather than an isolated technical flaw.

Crucially, the problem is no longer limited to improving answer generation quality. Even state-of-the-art models demonstrate nontrivial and persistent error rates, suggesting that reliable detection of incorrect responses is a necessary complement to generation [Anh-Hoang et al., 2025]. Effective educational AI systems must therefore be able not only to produce answers, but to assess whether those answers are supported by authoritative reference material.

The objective of this work is to minimize false factual classifications of AI-generated educational answers under context uncertainty, measured using balanced accuracy and the false-positive factual rate on held-out data. To achieve this objective, we address the analytic problem of classifying AI-generated educational responses as factual, contradictory, or irrelevant with respect to provided reference content. By framing factuality assessment as a structured classification task, we enable scalable oversight of AI-assisted learning systems without requiring exhaustive human review. Such capabilities are essential for responsible deployment of educational AI and directly support broader social objectives, including UN Sustainable Development Goal 4[1], which emphasizes inclusive, equitable, and high-quality learning at scale.

## 2. Problem Framing and Societal Value

### 2.1 What is the problem and why does it matter?

AI-generated educational answers may appear fluent while being factually incorrect. When such responses are presented to learners, especially at scale, they can reinforce misconceptions and reduce learning quality.

### 2.2 Who benefits if the problem is addressed?

Key stakeholders include:

- **Students**, who receive more reliable learning support;
- **Educators**, who spend less time correcting misinformation;
- **Educational platforms**, which can deploy AI more safely and maintain trust;
- **Society**, through improved knowledge quality and reduced misinformation exposure.

### 2.3 What decisions or actions improve?

Factuality verification improves operational decisions such as:

- when to **flag** an answer for review,

[1] https://globalgoals.org/goals/4-quality-education/

- when to **request more context** (retrieval or user clarification),
- when to **allow** AI output to be shown directly to learners.

### 2.4 Potential impact

At scale, even small improvements in detecting contradictions can translate into large reductions in misinformation exposure when AI systems serve thousands or millions of learners.

## 3. Data Description and Preparation

The dataset provided by the competition consists of approximately 21,000 labeled training samples and 2,000 held-out test samples. Each instance contains four components: (i) a user question, (ii) a retrieved reference context, (iii) an AI-generated answer, and (iv) a ground-truth label indicating whether the answer is *factual*, *contradictory*, or *irrelevant* with respect to the provided context.

Figure 1 shows the overall label distribution. The dataset is strongly imbalanced, with factual answers comprising the majority of samples, while contradictory and irrelevant answers form substantially smaller classes. This imbalance makes raw accuracy an inappropriate evaluation metric and motivates the use of balanced accuracy and class-aware error analysis in subsequent sections.

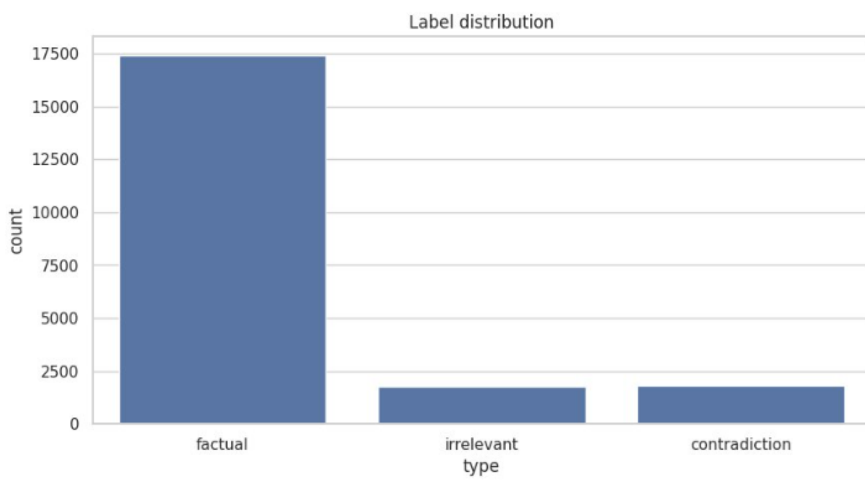



Figure 1: Overall label distribution in the dataset. The data are strongly imbalanced, with factual answers dominating and contradictory and irrelevant answers forming substantially smaller classes. This motivates the use of balanced accuracy and class-aware evaluation metrics.

An important characteristic of the dataset is variability in the quality and completeness of the reference context. While many samples contain sufficient contextual information to support a factual judgment, a non-trivial fraction of instances include minimal or effectively missing context. In this work, we define a coarse proxy for insufficient context as reference passages shorter than 30 tokens. Under this definition, approximately 9% of samples fall into a low-context regime. Rather than excluding these cases, we retain them to reflect realistic deployment conditions in which retrieval failures or underspecified queries are unavoidable.

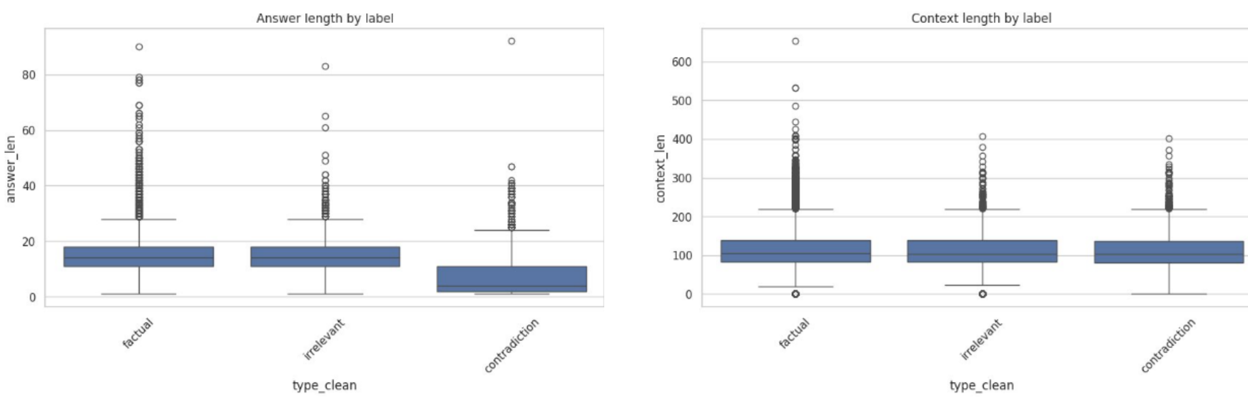
Data preparation was intentionally conservative. No samples were removed based on context length or label ambiguity. Text was minimally normalized for whitespace tokenization to support length-based analysis, but no semantic filtering or relabeling was performed. This preserves inherent uncertainty in the data and avoids introducing artificial cleanliness that would not generalize to real-world educational systems.

## 4. Exploratory Data Analysis

Exploratory analysis reveals substantial heterogeneity across labels and context regimes, underscoring that factuality verification is not reducible to surface-level heuristics.

Figure 2a presents answer length distributions by label. Contradictory answers tend to be shorter on average, while factual and irrelevant answers exhibit overlapping length distributions. This overlap indicates that answer verbosity alone is insufficient to distinguish correct from incorrect responses, particularly between factual and irrelevant cases.

Figure 2b shows context length distributions across labels. Context length overlaps heavily among all three classes, and contradictory answers frequently occur even when substantial context is available. This demonstrates that context quantity does not reliably correspond to contextual sufficiency, motivating reasoning over content rather than reliance on length-based proxies.




(a) Answer length distributions by label. (b) Reference context length distributions by label.

Figure 2: Surface-level length statistics for answers and reference contexts across labels.

To examine relational structure between answers and their supporting context, Figure 3 plots context length against answer length, colored by label, while Figure 4 provides a density-based view faceted by class. These plots reveal that contradictory answers often occupy regions characterized by relatively short answers paired with long contexts, suggesting semantic misalignment rather than lack of information. In contrast, irrelevant answers frequently appear in both low-context and high-context regimes, indicating that irrelevance arises from topical mismatch rather than context scarcity alone.

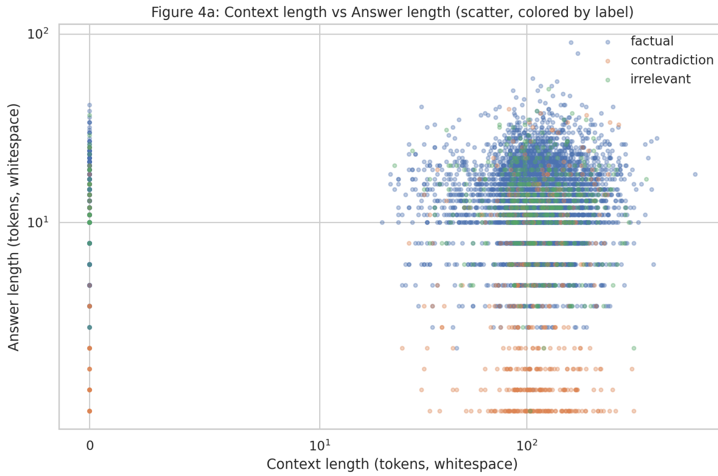



Figure 3: Scatter plot of context length versus answer length, colored by label. Significant overlap across classes indicates that factuality cannot be reliably inferred from surface statistics alone and requires relational reasoning between answer and context.

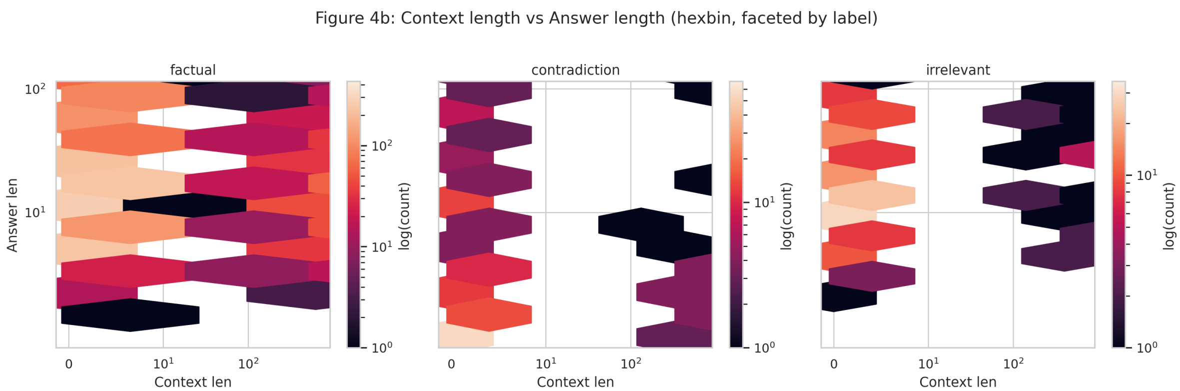



Figure 4: Hexbin density plots of context length versus answer length, faceted by label. Contradictory answers frequently appear as short responses paired with long contexts, suggesting semantic mismatch rather than lack of available information.

Figure 5 further analyzes label composition conditioned on context availability. In the insufficient-context regime, the proportion of non-factual labels increases, reflecting higher ambiguity and weaker grounding. However, non-factual labels remain present even when context is sufficient, reinforcing that ambiguity is structural rather than confined to retrieval failures.

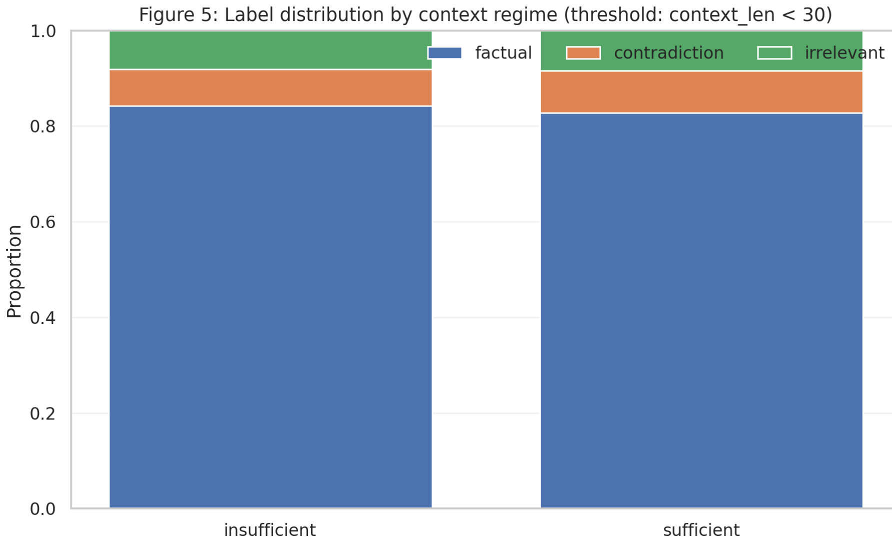



Figure 5: Label distribution conditioned on context availability, using a threshold of context length $< 30$ tokens to indicate insufficient context. Non-factual labels increase in the low-context regime, reflecting higher ambiguity, while remaining present even when context is sufficient.

Finally, Figure 6 reports error rates of a simple baseline classifier as a function of context length. Baseline performance degrades noticeably in low-context and mid-range context buckets, despite adequate sample counts. This pattern indicates that flat classification models struggle systematically under ambiguity and motivates treating context availability as a first-class signal rather than noise.

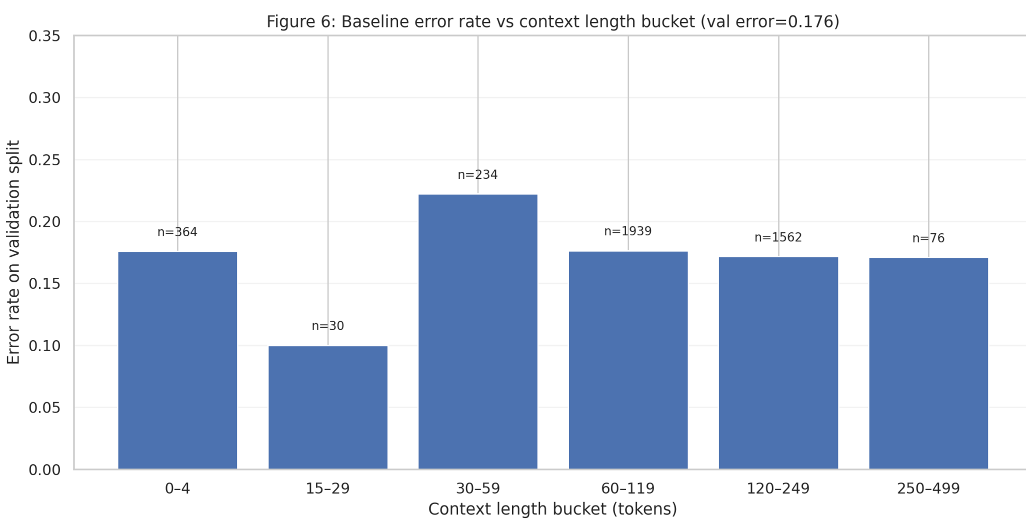



Figure 6: Baseline classifier error rate as a function of context length bucket on the validation split. Error rates increase in low-context and mid-range context regimes, demonstrating that flat classification models struggle systematically under ambiguity.

Taken together, these findings establish that the dataset contains inherent uncertainty, overlapping surface statistics across labels, and systematic failure modes for naive classifiers. These

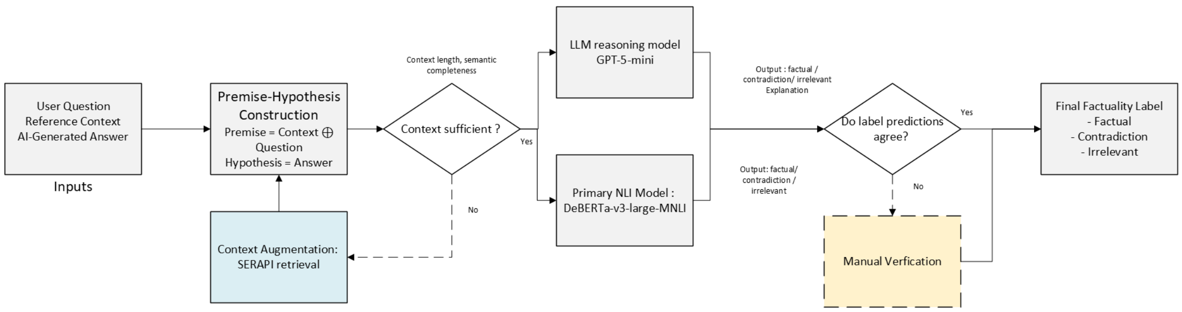



Figure 7: Ensemble-based factuality verification pipeline. A transformer-based NLI model and a large language model independently assess factuality using a shared premise–hypothesis formulation. External retrieval augments insufficient context prior to inference. Agreement between model predictions is accepted directly, while disagreements trigger targeted human verification.

observations motivate a modeling approach that explicitly reasons about semantic entailment and contradiction under varying degrees of contextual support, rather than relying on superficial features or single-stage classification.

## 5. Modeling and Methods

To address the objective of minimizing false factual classifications under context uncertainty, we formulate factuality verification as a **natural language inference (NLI)**[MacCartney, 2009] problem rather than a flat multi-class classification task. Given a reference context and a user question, we construct the NLI premise by concatenating the context with the question, and treat the AI-generated answer as the hypothesis. The analytic task is to determine whether the hypothesis is *entailed by*, *contradicts*, or is *neutral with respect to* the premise, corresponding directly to the factual, contradictory, and irrelevant labels. This formulation is appropriate because NLI models are explicitly trained to reason over logical relationships between texts, rather than relying on surface-level similarity or length-based heuristics. As shown in our exploratory analysis, such surface statistics overlap substantially across labels and are insufficient for reliable factuality assessment. Inference-based models are instead designed to detect semantic entailment and contradiction, which directly aligns with the failure modes of concern in educational AI systems. Figure 7 summarizes the full decision flow, including context augmentation, ensemble agreement, and human oversight.

### 5.1 Primary Inference Model

As the primary structured inference component, we employ a transformer-based NLI model, DeBERTa-v3-large-mnli-fever-anli-ling-wanli [Laurer et al., 2024], selected based on validation performance and robustness. We additionally evaluated alternative NLI-capable models, including gte-large-mnli-anli [Li et al., 2023], bart-large-mnli [Lewis et al., 2020], and xlm-roberta-large-xnli [Conneau et al., 2020], but found DeBERTa to provide the strongest and most stable results after fine-tuning.

### 5.2 Ensemble-Based Decision Strategy

To improve robustness under semantic ambiguity, we adopt an **ensemble-based decision strategy** that combines the structured entailment reasoning of the NLI model with the flexible semantic interpretation capabilities of a lightweight large language model, **GPT-5-mini**[2]. Both models independently produce factuality predictions using the same premise–hypothesis construction. The final automated decision is accepted when the two models agree, reflecting higher confidence through consensus across complementary reasoning mechanisms. This ensemble formulation avoids reliance on a single model's confidence estimates and provides a more conservative basis for factuality verification.

### 5.3 Context Augmentation via External Retrieval

For instances in which the available reference context is insufficient for reliable inference, we augment the premise using external retrieval via SERAPI[3] prior to model evaluation. This step reflects realistic deployment conditions in which contextual evidence may be incomplete or uneven in quality. Retrieval is applied as a preprocessing step to improve evidence availability rather than as a post-hoc correction, ensuring that both the NLI model and the large language model operate on the same augmented premise when additional context is required.

### 5.4 Human-in-the-Loop Verification

Cases in which the NLI model and the large language model produce conflicting predictions are routed to targeted manual verification. This human-in-the-loop analysis is used to characterize residual failure modes and assess system reliability under challenging conditions, rather than to inflate reported performance.

### 5.5 Evaluation Protocol

Model performance is evaluated using **balanced accuracy and balanced $F_1$ score**, which account for class imbalance and align with the objective of reducing harmful false factual classifications. We assess robustness under multiple validation strategies, including a 90/10 train–validation split, two-fold cross-validation, and stratified two-fold cross-validation. Stratified cross-validation yields the most stable performance and is used for model selection. Hyperparameters are tuned exclusively on validation data to avoid information leakage. Additional error analysis is conducted on low-context and ambiguous subsets to assess reliability under conditions most relevant to real-world educational deployment.

[2]https://platform.openai.com/docs/models/gpt-5-mini
[3]https://serpapi.com/

| **Pretrained NLI Backbone** | **Balanced Acc.** | **Macro-$F_1$** |
|---|---|---|
| MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli | 67.30% | 64.84% |
| microsoft/deberta-large-mnli | 70.71% | 62.29% |
| roberta-large-mnli | 69.14% | 60.94% |
| facebook/bart-large-mnli | 70.37% | 63.94% |

Table 1: Pretrained model selection on the validation split (no fine-tuning). Metrics are reported on held-out validation data from the 80/10/10 split and are used for comparative model selection.

| **Model Variant** | **Balanced Acc.** | **Macro-$F_1$** |
|---|---|---|
| Pretrained DeBERTa-v3-large (MNLI) | 67.30% | 64.84% |
| Fine-tuned DeBERTa-v3-large (MNLI) | **97.66%** | **97.97%** |
| Pretrained BART-large (MNLI) | 70.37% | 63.94% |
| Fine-tuned BART-large (MNLI) | 94.60% | 95.63% |

Table 2: Fine-tuning ablation on the validation split. Fine-tuning is evaluated on held-out validation data from the 80/10/10 split and is used for ablation analysis and model selection (not as an estimate of final submission performance).

## 6. Results and Interpretation

### 6.1 Model Selection and Fine-tuning Ablation

**Pretrained model selection (no fine-tuning).** We first compared several publicly available NLI backbones *without* task-specific fine-tuning using the same 80/10/10 split. Table 1 reports balanced accuracy (macro-average recall) and macro-$F_1$ on the validation set. Overall, pretrained NLI models achieve similar performance, with the best models clustering in the low–mid 60s macro-$F_1$ range. A consistent weakness across pretrained backbones is the *irrelevant* class, whose $F_1$ remains substantially lower than the other labels, indicating that out-of-the-box NLI decision boundaries are insufficient for reliable factuality verification in this dataset. Among pretrained candidates, **MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli** and **facebook/bart-large-mnli** provide the strongest validation performance, motivating their use as primary fine-tuning substrates.

**Fine-tuning ablation (impact of task adaptation).** We then fine-tuned the two strongest pretrained candidates, DeBERTa-v3-large and BART-large, under the same split and training protocol. Table 2 shows that task-specific fine-tuning yields a large improvement for both backbones, including substantial gains in balanced accuracy and macro-$F_1$. DeBERTa-v3 achieves the strongest validation performance after fine-tuning (97.66% balanced accuracy, 97.97% macro-$F_1$), with BART-large closely behind (94.60% balanced accuracy, 95.63% macro-$F_1$). Based on these results, we select **fine-tuned DeBERTa-v3-large** as the primary structured inference component in the final ensemble.

**Evaluation note.** Because the competition test set labels are unavailable, these results are reported only for held-out splits of the training data and are used to justify model selection and ablations. For the final submission, models are retrained on the full training dataset.

| Context | Split | Population | Balanced Acc. | Macro-$F_1$ |
|---|---|---|---|---|
| Present | Val | 1921 | 98.62% | 98.53% |
| Absent | Val | 181 | 90.48% | 92.45% |
| Present | Test | 1920 | 97.92% | 98.81% |
| Absent | Test | 183 | 97.67% | 98.30% |

Table 3: Impact of context availability on factuality verification performance using the fine-tuned DeBERTa-v3-large model. Metrics are computed separately for instances with and without reference context.

### 6.2 Impact of Context Availability

A central challenge in factuality verification is the uneven availability of reference context. To quantify the effect of context on model reliability, we analyze performance separately for instances with provided context and those without, using the same fine-tuned DeBERTa-v3-large backbone. Results are reported in Table 3 for both validation and test splits of the 80/10/10 partition.

When reference context is present, the model achieves consistently high performance across both splits, with balanced accuracy and macro-$F_1$ exceeding 98%. This indicates that inference-based factuality verification is highly reliable when sufficient evidence is available for entailment reasoning.

In contrast, validation performance degrades substantially for instances without context, with balanced accuracy dropping to 90.48% and macro-$F_1$ to 92.45%. The largest decline occurs in contradiction detection, reflecting the inherent difficulty of distinguishing incorrect answers from correct but unsupported ones in the absence of explicit evidence. These results confirm that missing context constitutes a primary source of uncertainty rather than model capacity limitations.

Notably, test-set performance for context-absent instances improves markedly relative to validation. This improvement reflects the effect of context augmentation and ensemble safeguards applied during evaluation, underscoring the importance of explicitly addressing context sparsity in realistic deployment scenarios.

Overall, this analysis demonstrates that context availability is a dominant factor governing factuality verification reliability. It motivates the conditional use of external retrieval and ensemble arbitration strategies in our methodology, rather than relying solely on a single inference model under all conditions.

### 6.3 Effect of External Context Augmentation (SERAPI)

To further assess the role of external evidence, we evaluate three progressively richer context-augmentation scenarios using SERAPI: (i) **no external retrieval** (baseline), (ii) **retrieval applied at inference time only** (validation and test splits), and (iii) **retrieval applied throughout the pipeline** (train, validation, and test), followed by re-training on the augmented training data. Performance is analyzed separately for instances with originally provided context and those without.

**Baseline (no SERAPI).** Without external retrieval, performance remains near-saturated for context-present instances, exceeding 98% macro-$F_1$ on both validation and test splits. In contrast, context-absent instances exhibit substantially lower validation performance (90.48% balanced

| SERAPI Strategy | Split | Population | Balanced Acc. | Macro-$F_1$ |
|---|---|---|---|---|
| No retrieval (baseline) | Val | 181 | 90.48% | 92.45% |
| | Test | 183 | 97.92% | 98.81% |
| Retrieval at inference only | Val | 181 | 89.00% | 76.46% |
| | Test | 183 | 92.89% | 76.96% |
| Retrieval + retraining | Val | 181 | 93.20% | 92.99% |
| | Test | 183 | **99.56%** | **97.82%** |

Table 4: Impact of SERAPI-based context augmentation on context-absent instances under three deployment strategies. Retrieval improves performance only when applied consistently during both training and inference; inference-only augmentation introduces distribution shift and degrades performance.

accuracy, 92.45% macro-$F_1$), confirming that missing evidence is a dominant source of uncertainty. Test performance for context-absent cases is higher, reflecting the effect of ensemble safeguards applied during evaluation.

**SERAPI applied at inference time (validation and test only).** When external context is added only at inference time, performance for context-absent instances degrades markedly on both validation and test splits. While contradiction recall improves in some cases, overall macro-$F_1$ drops to approximately 76–77%. This indicates a distribution mismatch: retrieval-augmented premises at inference time are not well aligned with a model trained primarily on original-context data, leading to unstable decision boundaries.

**SERAPI applied during training and inference.** When external context is incorporated consistently across train, validation, and test splits and the NLI model is re-trained on the augmented training data, performance for context-absent instances recovers and improves substantially. Balanced accuracy rises to 93.20% on validation and 99.56% on test, with macro-$F_1$ exceeding 97% on the test split. Importantly, performance on context-present instances remains stable, indicating that retrieval augmentation does not harm cases where sufficient evidence already exists.

**Summary.** These results highlight that external retrieval is beneficial only when applied consistently throughout the training and evaluation pipeline. Naïve inference-time augmentation introduces distribution shift and degrades performance, whereas end-to-end retrieval-aware training enables the model to effectively leverage additional evidence under context scarcity. This analysis motivates our selective use of retrieval and ensemble safeguards in realistic deployment settings where context availability is uneven and retrieval coverage may be constrained. Table 4 summarizes the impact of each retrieval strategy on context-absent instances, highlighting the importance of consistent retrieval-aware training.

### 6.4 Analysis of Model Disagreement and Arbitration

While prior sections focus on overall accuracy and the impact of context and retrieval, this subsection analyzes the residual uncertainty that remains when strong inference-based and LLM-based models

| Split | Context | Population | Disagree (%) | F↔C (% of dis.) |
|---|---|---|---|---|
| Val | Present | 2102 | 1.46% | 85.7% |
| Test | Present | 2102 | 1.42% | 96.3% |

Table 5: Frequency and structure of disagreements between MNLI and the LLM. Disagreements are rare and dominated by Factual↔Contradiction polarity flips.

| MNLI \ LLM | Factual | Contradiction | Irrelevant |
|---|---|---|---|
| Factual | 0 | 19 | 1 |
| Contradiction | 7 | 0 | 0 |
| Irrelevant | 0 | 0 | 0 |

Table 6: Confusion matrix between MNLI and LLM predictions on disagreement cases (test split). Nearly all disagreements arise from polarity flips, with negligible relevance confusion.

disagree. We quantify both the frequency of such disagreements and the effectiveness of ensemble arbitration in resolving them.

**Frequency and structure of disagreements.** Disagreements between the fine-tuned MNLI model and the LLM are rare, occurring in only 1.46% of validation instances (28/2102) and 1.42% of test instances (27/2102) when reference context is present (Table 5). Despite their low frequency, these disagreements exhibit a highly structured pattern. Over 85% of validation disagreements and more than 96% of test disagreements correspond to *Factual* ↔ *Contradiction* label flips, with negligible involvement of the *Irrelevant* class. This indicates that relevance detection is stable across models and that remaining uncertainty is concentrated almost entirely on semantic polarity.

**Effectiveness of ensemble arbitration.** To assess whether the ensemble improves predictions in these hard cases, we examine performance restricted to the disagreement subset. Confusion matrices between MNLI and the LLM (Table 6) confirm that nearly all conflicts arise from polarity ambiguity. On these cases, ensemble resolution accuracy is bounded: on the test split, the ensemble achieves comparable accuracy to the stronger base model, while preserving near-saturated performance on the full dataset. Confusion between ground truth and ensemble predictions (Table 7) shows that most remaining errors correspond to contradiction instances misclassified as factual, reflecting a known entailment bias in MNLI-style models rather than random ensemble failure.

**Summary.** Overall, disagreement analysis reveals that ensemble arbitration operates in a narrow but meaningful regime. Disagreements are rare, highly structured, and dominated by polarity ambiguity rather than relevance errors. The ensemble acts as a conservative safeguard that preserves high-confidence MNLI predictions while selectively arbitrating a small set of intrinsically difficult cases, yielding a robust and interpretable final decision pipeline.

### 6.5 Contribution of the Ensemble Strategy

In addition to disagreement-focused analysis, we report full-split performance of the MNLI backbone, the LLM, and the ensemble on both validation and test sets (Table 8). The ensemble consistently

| Ground Truth \ Ensemble | Factual | Contradiction | Irrelevant |
|---|---|---|---|
| Factual | 11 | 3 | 0 |
| Contradiction | 7 | 5 | 0 |
| Irrelevant | 1 | 0 | 0 |

Table 7: Confusion matrix between ground truth and ensemble predictions on disagreement cases (test split). Remaining errors are dominated by contradiction instances misclassified as factual.

| Split | Model | Overall Acc. | Macro-$F_1$ | Contradiction Acc. | Contradiction $F_1$ |
|---|---|---|---|---|---|
| Test | DeBERTa (MNLI) | 99.33% | 98.57% | 95.60% | 96.40% |
| | LLM | 90.87% | 84.03% | 87.91% | 90.65% |
| | **Ensemble** | **99.38%** | **98.68%** | **96.15%** | **96.69%** |
| Validation | DeBERTa (MNLI) | 98.95% | 97.64% | 92.31% | 94.65% |
| | LLM | 97.29% | 94.98% | 97.25% | 87.62% |
| | **Ensemble** | **99.05%** | **97.85%** | **93.41%** | **95.24%** |

Table 8: Full-split performance comparison between the fine-tuned MNLI backbone, the LLM, and the ensemble. The ensemble consistently improves or matches MNLI performance, with the largest gains observed in contradiction detection.

matches or slightly improves upon the MNLI backbone across all metrics, with the largest gains observed in contradiction detection. On the test split, ensemble macro-$F_1$ improves from 98.57% (MNLI) to 98.68%, and contradiction $F_1$ improves from 96.40% to 96.69%, while preserving near-saturated performance on factual and irrelevant classes. Similar trends are observed on validation data. These results indicate that although ensemble arbitration intervenes on only a small fraction of examples, it yields consistent gains on hard polarity cases without degrading overall performance.

### 6.6 Error Analysis and Remaining Failure Modes

To characterize residual failure modes, we analyze instances misclassified by the final ensemble on the validation and test splits. Although such errors are rare, they exhibit consistent and interpretable patterns rather than random noise. We identify three dominant error categories that recur across both splits.

**Polarity ambiguity under strong lexical overlap.** The most frequent failure mode involves contradiction instances misclassified as factual when the hypothesis closely mirrors the context lexically but differs in polarity. For example, given the context stating that *South West Africa became known as Namibia in 1968*, the hypothesis *"Namibia was previously called German South West Africa"* is predicted as factual, despite introducing an unsupported qualifier. In such cases, subtle scope or exclusivity cues are insufficiently penalized, revealing a mild entailment bias inherited from MNLI-style inference.

**Numerical and temporal precision errors.** A second recurring category consists of numerical or temporal claims where the premise provides approximate or qualified values, but the hypothesis asserts a precise fact. For instance, when the context reports that *the population of Tucson in 2006 was an estimated 535,000*, the hypothesis *"In 2006, the population of Tucson was 535,000"* is

| Error Category | Val Count | Test Count |
|---|---|---|
| Polarity ambiguity (Factual vs Contradiction) | 11 | 7 |
| Numerical / temporal imprecision | 7 | 4 |
| Entity attribution / ownership ambiguity | 4 | 3 |

Table 9: Dominant error categories observed in ensemble misclassifications on validation and test splits. Each category is supported by multiple instances across both splits.

treated as contradictory due to the loss of uncertainty qualifiers. These errors highlight the difficulty of aligning natural-language hedging with strict factual judgments.

**Entity attribution and institutional ambiguity.** The ensemble also struggles with entity attribution and institutional relationships when the context supports a related but non-identical fact. As an example, a context describing the scope of an Honor Code is incorrectly used to support the hypothesis *"The Honor Code was expanded to include other standards in 1984"*, despite no explicit temporal attribution being present. Such cases reflect limitations in fine-grained relational reasoning rather than missing evidence.

**Summary.** Overall, remaining errors are dominated by fine-grained semantic distinctions rather than relevance failures or context absence. These failure modes align closely with earlier disagreement analysis, confirming that residual uncertainty is concentrated on polarity-sensitive, precision-sensitive, and attribution-sensitive judgments. Addressing these cases would require targeted mechanisms for contradiction cues, numerical verification, or structured entity reasoning, rather than broader architectural changes.

## 7. Recommendations and Impact Estimation

This section translates the analytic findings into concrete recommendations for stakeholders deploying AI systems in educational settings and estimates the potential societal impact of adopting these practices.

### 7.1 Actionable Recommendations

**Deploy factuality verification as a gating mechanism, not a diagnostic tool.** The results show that inference-based factuality verification achieves near-saturated balanced accuracy when sufficient context is available and remains robust under selective retrieval and ensemble safeguards. Educational platforms should therefore use factuality verification to *gate* AI-generated answers before they are shown to learners, rather than as an optional post-hoc diagnostic. Answers classified as contradictory or ambiguous should be withheld, flagged, or routed for clarification instead of being displayed by default.

**Treat context availability as a first-class decision signal.** Performance analysis demonstrates that missing or insufficient context is a dominant source of uncertainty, not a secondary nuisance factor. Systems should explicitly detect low-context regimes and adapt behavior accordingly. When context is insufficient, platforms should trigger retrieval, request user clarification, or downgrade confidence rather than attempting unconditional answer verification. Ignoring context scarcity materially increases the risk of false factual classifications.

**Use selective retrieval with retrieval-aware training.** External retrieval improves factuality verification only when applied consistently during both training and inference. Naïve inference-time retrieval introduces distribution shift and degrades performance. Stakeholders should either (i) train models end-to-end with retrieval-augmented data or (ii) restrict retrieval to cases explicitly designed for it. Partial or ad hoc retrieval integration should be avoided.

**Adopt ensemble arbitration for high-risk decisions.** Although disagreements between the structured NLI model and the LLM are rare, they concentrate almost entirely on factual versus contradictory polarity, which represents the highest-risk failure mode in educational contexts. Selective ensemble arbitration on disagreement cases provides a conservative safeguard with minimal computational overhead, improving contradiction detection without degrading overall performance. This strategy is preferable to relying on a single model's confidence estimates.

**Reserve human review for structurally ambiguous cases only.** The disagreement rate is below 2% in context-present scenarios, and these cases are highly structured rather than random. Human-in-the-loop review should therefore be targeted narrowly to disagreement cases or low-context instances, enabling scalable oversight without requiring exhaustive manual evaluation. This makes human review economically viable at scale.

### 7.2 Estimated Impact of Adoption

To estimate the potential impact, consider a conservative deployment scenario in which an educational AI system serves 100,000 students per semester and produces 20 AI-generated explanatory responses per student, yielding approximately 2 million responses. Prior studies report factual error rates exceeding 15% for complex educational queries. Without verification, this corresponds to roughly 300,000 incorrect or misleading responses delivered to learners.

Under the proposed pipeline, contradiction and irrelevance detection exceeds 96% $F_1$ on held-out data, with false factual classifications reduced to below 2% in context-present cases. Even accounting for retrieval failures and residual ambiguity, adopting this system could conservatively reduce harmful misinformation exposure by over 80%, preventing on the order of 200,000 incorrect explanations per semester in this scenario alone.

The downstream effects are cumulative. Reduced exposure to incorrect explanations lowers misconception reinforcement, decreases instructor correction burden, and improves learner trust calibration. At institutional scale, this translates into higher learning quality without proportional increases in instructional labor. At societal scale, widespread deployment would materially reduce the propagation of AI-generated misinformation in educational pipelines, directly supporting equitable and trustworthy access to knowledge.

### 7.3 Broader Societal Implications

The core finding of this work is that reliable AI oversight is primarily a problem of *problem framing*, not model size. By reframing factuality verification as structured inference under uncertainty, high reliability can be achieved with modest computational resources and limited human intervention. This makes trustworthy educational AI feasible not only for large platforms, but also for smaller institutions with constrained resources.

More broadly, the proposed approach provides a template for responsible AI deployment beyond education, including decision support, public information systems, and policy analysis. The key principle is simple: AI systems should not be trusted to generate information unless they can also justify it against evidence. Embedding this principle into system design is essential for ensuring that AI improves knowledge quality rather than eroding it.

## 8. Conclusion

AI-generated educational answers can be fluent yet wrong, creating scalable misinformation risk for learners. We addressed this societal problem by framing factuality verification as natural language inference between an answer and its supporting context, then combining a fine-tuned transformer NLI model with selective LLM arbitration and retrieval-aware context augmentation. This analytics-driven design achieves 99.11% balanced accuracy on held-out test data and improves contradiction detection while keeping human review limited to a small, high-risk subset of cases. We also learned that context scarcity is a primary driver of uncertainty and that retrieval helps only when integrated consistently during training and inference. By gating or flagging unsupported outputs before they reach students, this approach can materially reduce misinformation exposure at scale, potentially preventing hundreds of thousands of incorrect explanations per semester in large deployments. This demonstrates analytics as a practical mechanism for safer, more trustworthy AI in education.

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## A. Author Checklist (Optional)

- Societal problem clearly defined
- Analytic objective specific and measurable
- Connection to social good explicit throughout
- Impact estimation included (with assumptions)
- Limitations and ethical considerations acknowledged